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LABELS AND ANNOTATIONS

Files

5G AND 6G LAB

LAB EXPERIMENT

New Folder

ASSEMENT QNO 1.m

EX: 2.m

ex 1.m

EX 2 ..m

EX 2..m

Ex 2.m

ex 5.m

ex1.m

Workspace

Name	Value	Size	Class
Capacity	4×16 double	4×16	double
Gain	128	1×1	double
i	4	1×1	double
j	16	1×1	double
Nt	[8,32,64,128]	1×4	double
SNR	1×16 double	1×16	double
SNRdB	1×16 double	1×16	double

EX4.m

untitled4.m

EX6.m

EX7.m

EX8.m

EX9.m

1

2

3

4

5

6

7

8

9

10

11

12

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```

clc; clear; close all;

% Massive MIMO Capacity vs SNR and Gain

Nt = [8 32 64 128]; % Number of Base Station Antennas
SNRdB = 0:2:30; % SNR Range (dB)
SNR = 10.^(SNRdB/10);

Capacity = zeros(length(Nt), length(SNR));

for i = 1:length(Nt)

    Gain = Nt(i); % Massive MIMO Array Gain

    for j = 1:length(SNR)

        % Shannon Capacity with Array Gain
        Capacity(i,j) = log2(1 + Gain*SNR(j));

    end
end

```

Figure 1

Massive MIMO Capacity vs SNR and Array Gain

SNR (dB)	8 Antennas	32 Antennas	64 Antennas	128 Antennas
0	3.2	4.8	6.4	8.0
5	4.0	5.6	7.2	8.8
10	4.8	6.4	8.0	9.6
15	5.6	7.2	8.8	10.4
20	6.4	8.0	9.6	11.2
25	7.2	8.8	10.4	12.0
30	8.0	9.6	11.2	12.8

Command Window

>>